

# Halvor Hafnor

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## SUMMARY

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Embedded software and machine learning engineer building classification pipelines for real sensor data on constrained hardware, from collection through feature extraction to on-device inference. Two years at Aerendir Mobile, a shipped automotive product, a pending patent, and an M.S. at UC San Diego.

## EXPERIENCE

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### Embedded ML Engineer

*Aerendir Mobile*

Jul. 2024 – Present

*Mountain View, CA*

- Developed embedded machine learning pipelines for classifying biometric motion signals from inertial sensors on **Linux**-based mobile devices.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the motion dataset behind the company's biometric authentication work: **400 participants**, each recorded holding the phone in three postures, sitting, standing and walking, from the accelerometer and gyroscope.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel**. Evaluated ST's on-sensor **Machine Learning Core**, then replaced it with a threshold classifier over hand-engineered features from a **240 Hz** six-axis IMU, running in the sensor's own firmware. Better than **99% accuracy** at under **0.5% false positives**, answering within **1.5 seconds**. Subject of a pending patent naming me as an inventor.

### Electrical Engineering Intern

*ABB Electrification*

Jun. 2021 – Sep. 2023

*Skien, Norway*

- Assembled and wired **around 300** industrial control panels and power cabinets from engineering schematics, across three consecutive summers.
- Supported production and testing of custom electrical systems for industrial clients.

## EDUCATION

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### University of California - San Diego - San Diego, CA

*Computer Engineering, M.S.*

Jan. 2026 – June 2027

*Computer Engineering, B.S.* Major GPA: 3.9

Sep. 2024 – June 2026

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

### De Anza College - Cupertino, CA

*Computer Science & Engineering*

GPA: 4.0

*Academic and Technical Roles*

Sep. 2022 – June 2024

- Math & Science Tutor and CS Teaching Assistant for calculus, physics, discrete math, C++, Java and Assembly.

## TECHNICAL SKILLS

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**Languages:** C++, C, Python, TypeScript, Java, Bash, ARM Assembly

**Embedded:** Embedded Linux, Android, QML/Qt, inertial sensing, on-sensor inference, ST IMU firmware, Docker

**Signals & ML:** Signal processing, feature extraction, classification, OpenCV, NumPy, SciPy, YOLOv8

**Systems:** Node, SQLite, WebSocket protocols

**Agentic Development:** Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

## PROJECTS

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**TabTerm**, a macOS terminal in Chrome tabs

- A **Chrome extension** over a background **terminal daemon** that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search and project-scoped saved commands.

**Laser targeting on an autonomous car**, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5, in **Docker**. **YOLOv8** detection and **OAK-D** stereo depth, converted through a **71.9°** field of view into pan and tilt angles so two servos hold a laser on the nearest person from **0.5 to 2.9 m**. Depth is the median of a 10x10 patch, since one bad pixel throws the aim.